



Application Lifecycle Management

Software Version: 14.0x

Microsoft Excel Add-in Guide

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Welcome to This Guide

Welcome to the Application Lifecycle Management (ALM) Microsoft Excel Add-in, which enables you to upload requirements, tests, and defects from Microsoft Excel directly to ALM.

This guide explains how to use Microsoft Excel with ALM. For additional information on using ALM, refer to the *Micro Focus Application Lifecycle Management User Guide*.

How This Guide is Organized

This guide contains the following chapters:

Chapter	Description
"Installing the ALM Microsoft Excel Add-in" on page 5	Describes how to install the Microsoft Excel Add-in.
"Formatting Data" on page 9	Describes how to format requirement, test plan, or defect data in a Microsoft Excel worksheet.
"Uploading Data to ALM" on page 15	Describes how to upload the Microsoft Excel worksheet to ALM.
"Limitations" on page 20	Describes the limitations when working with the ALM Microsoft Excel Add-in.

Chapter 1: Installing the ALM Microsoft Excel Add-in

To upload requirements, test plans, or defects from Microsoft Excel to ALM, you must install the ALM Microsoft Excel Add-in on your client machine.

Before you install the ALM Microsoft Excel Add-in, make sure that Microsoft Excel is installed on your machine.

Note: To install this add-in for all users, you must log in with administrator privileges.

To install the ALM Microsoft Excel Add-in:

1. Close Microsoft Excel.
2. Uninstall any previous versions of the ALM Microsoft Excel Add-in. To uninstall, choose **Start > Control Panel > Programs > Programs and Features** and follow the instructions on your screen.
3. Download and install the add-in from [Marketplace](#). Follow the on-screen installation instructions.
4. After you install the Excel Add-in, configure your Microsoft Excel security settings. For more information on configuring Excel security settings, refer to the Microsoft Excel documentation.
5. Before you use the Excel Add-in for the first time on your machine, you must download the ALM client. Navigate to the ALM login screen, and the appropriate files are downloaded to your machine.

Chapter 2: Understanding the ALM Upload Tab

After installing the ALM Microsoft Excel Add-in, a new tab, the ALM Upload Add-in, is added to Excel.

The tab contains the following options:

Field	Description
Login	Opens the login screen to log in to ALM. See "Logging in to ALM" on page 17 for details.
Domain	Select a new domain in ALM to update with your Excel data.
Project	Select a new project in ALM to update with your Excel data.
Open	Opens the file explorer to search for existing mapping files to use for mapping your Excel data. See "Mapping Your Data" on page 17 for details.
Mapping	Opens the Field Mapping dialog box. See "Mapping Your Data" on page 17 for details.
Validate	Validates your Excel data. See "Validating Your Data" on page 18 for details.
Upload to ALM	Uploads your data to the ALM project that you are logged into.
Help	Opens this document.

Chapter 3: Formatting Data

Before you can upload requirements, tests, and defects from Excel to ALM, you must format the data. This chapter explains how to format the data in Microsoft Excel.

This chapter includes:

• Guidelines for Formatting	10
• Formatting Requirements	11
• Formatting Tests	12
• Formatting Defects	13

Guidelines for Formatting

When formatting requirements, tests, or defects in your Excel worksheet, consider the following:

- Arrange the records that you want to upload in rows.
- For each field that you want to upload, designate a column. Make sure to include all the required fields for requirements, tests, and defects, as defined in your ALM project.
- You can manually map ALM fields to worksheet columns, or the Microsoft Excel Add-in can automatically map them. To enable automatic mapping:
 - Include a header row with heading names that are identical to the ALM field names.
 - The automatic mapping scans the header row for matching fields until the first empty cell. Make sure not to leave empty cells in the header row.
- Make sure that each requirement, test, or defect contains data for all the required fields, as defined in your ALM project. For details about the system and user fields defined in your project, see the information about customizing projects in the *Micro Focus Application Lifecycle Management Administrator Guide*.
- To upload multiple values to ALM fields that support multiple values, separate the values with a semicolon (;) character.
- Field data must not exceed the field size in ALM.
- Data intended for ALM fields that have selection lists must match the entries in those selection lists exactly. For example, the **Priority** field can accept only one of the following values: **5-Urgent**, **4-Very High**, **3-High**, **2-Medium**, or **1-Low**.
- Only plain text and hyperlinks are uploaded to ALM. Any graphics are ignored, and formatting is removed.
- You can upload attachments for requirements, tests, or defects. For more information, see ["Formatting Attachments" on the next page](#).
- You can upload design steps for tests.
- You cannot upload information indirectly related to requirements, tests, or defects. For example, if uploading tests, you cannot specify the requirements covered by each test.

However, if you modify or add tests in Excel and upload the data again to ALM, the existing tests are updated in the test plan module, and any test coverage to requirements is preserved.

- Each requirement, test, or design step on the same hierarchical level must have a unique name. If a name is not unique, it will be overwritten. For example, if your worksheet contains two requirements on the same hierarchical level with the same name, only one is created.

For additional guidelines on formatting requirements, tests, or defects, see ["Formatting Requirements" on the next page](#), ["Formatting Tests" on page 12](#), and ["Formatting Defects" on page 13](#).

Formatting Attachments

When formatting attachments for requirements, tests, or defects, consider the following:

- Designate a separate column for attachments.
- To include attachments for test design steps, designate a separate column for design step attachments.
- An attachment can be either a **File** or a **URL**.
 - To upload a file, specify the file name with its full path. For example, C:\file1.txt. Alternatively, type a hyperlink in the cell that points to the file.
 - To upload a URL, save the URL in a shortcut file, and then specify the full file path. You cannot enter a URL directly in the worksheet.
- To include multiple attachments, list the attachments in the same cell separated by semicolons (;).
Example: Attachment1;Attachment2;Attachment3

Formatting Requirements

Before you upload your requirements to the ALM Requirements module, you format the requirements in your worksheet.

To format a requirement:

1. Open your Excel worksheet.
2. Define each requirement on a separate row. For each requirement, you must define all required fields as defined in your ALM project.

Consider the following points when formatting requirements:

- **Requirement types.** You can assign a single requirement type to your requirements, or your requirements can include multiple requirement types. If they include multiple types:
 - Designate a **Requirement Type** column. For each requirement, specify the requirement type.
 - For each requirement, include data for the required fields of the specific requirement type.
- To create a child requirement, add a **Path** column to the Excel worksheet. In the **Path** column, type the path under which the requirement is created. Separate each hierarchical level with a backslash (\) character. Make sure not to use a forward slash. If the hierarchy does not exist in ALM, the hierarchy is created along with the requirements.
- If you enter data in a **Target Cycle** or **Target Release** field, you must include the path preceding the cycle or release. Separate each hierarchical level with a backslash (\) character. For example: parent1\parent2\TC1. Make sure not to use a forward slash. If a Target Release or Target Cycle value is unique, you do not need to include its preceding hierarchy.
- If you enter data in a **Target Cycle** field, you must also enter data in the **Target Release** field.

The following example illustrates these points.

	A	B	C	D	E	F
1	Name	Requirement Type	Author	Priority	Product	Path
2	Requirement_1	Business	alex_alm	5-Urgent	Mercury Tours Web Site	
3	Requirement_2	Functional	shelly_alm	3-High	Mercury Tours Web Site	Requirement_1
4	Requirement_3	Testing	shelly_alm			Requirement_1
5	Requirement_4	Testing	shelly_alm			Requirement_1\Requirement_3

The worksheet contains data columns for the **Name**, **Requirement Type**, **Author**, **Priority**, and **Product** fields.

The **Path** column does not correlate to an ALM field. It is used to indicate the hierarchical level of each requirement. For example, *Requirement_1* does not have an entry in that column, indicating that it is on the top hierarchical level. The path defined for *Requirement_2* and *Requirement_3* indicates that these requirements are child requirements of *Requirement_1*. The path defined for *Requirement_4* indicates that it is a child of *Requirement_3*.

Requirement_3 and *Requirement_4* are defined as the **Testing** requirement type and do not have entries in the **Priority** and **Product** columns. **Priority** and **Product** are not required fields for the **Testing** requirement type.

After you upload the requirements to ALM, *Requirement_1* is displayed under the root requirement, and the other requirements are displayed on other hierarchical levels below *Requirement_1*. *Requirement_3* and *Requirement_4* do not have entries in the **Priority** and **Product** fields.

Requirements Edit View Favorites Analysis					
No Filter Defined					
	Name	Requireme...	Author	Priority	Product
Requirements	Folder				
Requirement_1	Business	alex_alm	5-Urgent	Mercury Tours Web	
Requirement_2	Functional	shelly_alm	3-High	Mercury Tours Web	
Requirement_3	Testing	shelly_alm			
Requirement_4	Testing	shelly_alm			

For additional formatting guidelines, see ["Guidelines for Formatting" on page 10](#).

Once you have formatted your requirements, you can upload them to ALM. For more information, see ["Uploading Data to ALM" on page 15](#).

Formatting Tests

Before you upload your tests to the ALM Test Plan module, you format the tests in your worksheet.

To format a test:

1. Open your Excel worksheet.
2. Define each test on a separate row. For each test, you must define all required fields as defined in

your ALM project.

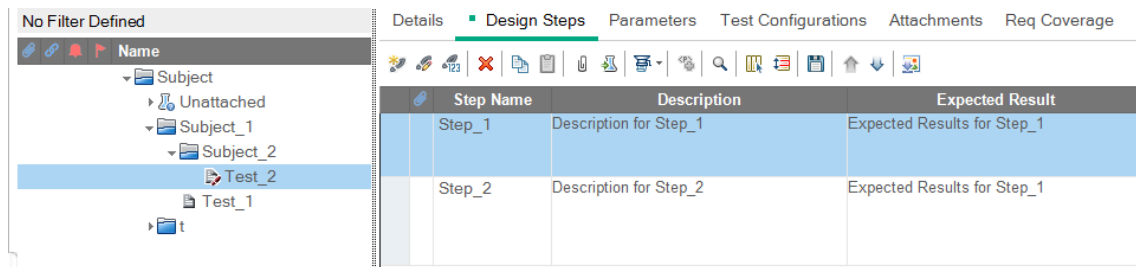
Consider the following points when formatting tests:

- **Subject.** Designate a **Subject** column, where you specify the test subject folder in which the test is created. To create tests in subfolders of the Test Plan tree, separate each hierarchical level with a backslash (\) character. Make sure not to use a forward slash.
- **Design steps.** Enter each design step on a separate row. Identify the test under which the design step is included by typing the **Subject** and **Test Name**.

The following example shows two tests. Each test contains two design steps. The **Subject** column indicates the hierarchical level of each subject test folder in the test plan tree. For example, *Subject_1* and *Subject_2* are on different hierarchical levels. *Subject_2* is a subfolder of *Subject_1*. *Subject_1* is placed directly beneath the pre-defined **Subject** root folder.

	A	B	C	D	E	F
1	Subject	Test Name	Description	Step Name	Step Description	Expected Results
2	Subject_1	Test_1	Description for Test_1	Step_1	Description for Step_1	Expected Results for Step_1
3	Subject_1	Test_1		Step_2	Description for Step_2	Expected Results for Step_2
4	Subject_1\Subject_2	Test_2	Description for Test_2	Step_1	Description for Step_1	Expected Results for Step_1
5	Subject_1\Subject_2	Test_2		Step_2	Description for Step_2	Expected Results for Step_2

After you upload the tests to ALM, *Subject_1* contains *Test_1* and *Subject_2* contains *Test_2*. The subject test folders are on different hierarchical levels.



The screenshot shows the ALM interface. On the left, a tree view under 'No Filter Defined' shows a hierarchy: Subject > Unattached > Subject_1 > Subject_2 > Test_2 (selected) > Test_1 > t. On the right, the 'Design Steps' tab is active, displaying a table with the following data:

Step Name	Description	Expected Result
Step_1	Description for Step_1	Expected Results for Step_1
Step_2	Description for Step_2	Expected Results for Step_1

For additional formatting guidelines, see ["Guidelines for Formatting" on page 10](#).

Once you have formatted your tests, you can upload them to ALM. For more information, see ["Uploading Data to ALM" on page 15](#).

Formatting Defects

Before you upload your defects to the ALM Defects module, you format the defects in your worksheet.

To format a defect:

1. Open your Excel worksheet.
2. Define each defect on a separate row. For each defect, you must define all required fields as defined in your ALM project.

Consider the following points when formatting defects:

- If you enter data in a **Detected in Release**, **Detected in Cycle**, **Target Release**, or **Target Cycle** field, you must include the path preceding the release or cycle. Separate each hierarchical level with a

backslash (\) character. For example: parent1\parent2\TC1. Make sure not to use a forward slash. If a **Target Release** or **Target Cycle** value is unique, you do not need to include its preceding hierarchy.

- If you enter data in a **Detected in Cycle** or **Target Cycle** field, you must also enter data in the **Detected in Release** or **Target Release** field.

For example, suppose you want to upload the following defects:

	A	B	C	D	E	F	G
1	Summary	Detected By	Detected on Date	Status	Subject	Severity	Priority
2	The list of flights is given v	alex_alm	28/02/2006	Open	Flight Reservation	3-High	3-High
3	The list of available flights	shelly_alm	28/02/2006	Open	Flight Reservation	5-Urgent	5-Urgent
4	The list of flights is given v	alex_alm	28/02/2006	Open	Flight Reservation	3-High	3-High
5	Nothing happened when p	shelly_alm	28/02/2006	Open	Flight Reservation	5-Urgent	5-Urgent

After you upload the defects to ALM, the defects are displayed in the Defects module.

	Defect ID	Summary	Detected By	Detected on...	Status	Subject	Severity	Priority
1		The list of flights is given...	alex_alm	06/08/2014	Open	Flight Reservation	3-High	3-High
2		The list of available flights	shelly_alm	06/08/2014	Open	Flight Reservation	5-Urgent	5-Urgent
3		The list of flights is given...	alex_alm	06/08/2014	Open	Flight Reservation	3-High	3-High
4		Nothing happened when...	shelly_alm	06/08/2014	Open	Flight Reservation	5-Urgent	5-Urgent

Note: When an ALM defect field has an input mask, the mask is not checked when you upload defects from Excel to ALM.

For additional formatting guidelines, see ["Guidelines for Formatting" on page 10](#).

Once you have formatted your defects, you can upload them to ALM. For more information, see ["Uploading Data to ALM" on page 15](#).

Chapter 4: Uploading Data to ALM

After you define requirements, tests, or defects in your worksheet, you can upload them to ALM.

This chapter includes:

• Considerations for Uploading Data to ALM	16
• Version Control Considerations	16
• Logging in to ALM	17
• Mapping Your Data	17
• Validating Your Data	18
• Uploading Data to ALM	18

Considerations for Uploading Data to ALM

- To upload data to ALM, you must first map ALM fields to the Excel worksheet columns. You can manually map ALM fields to worksheet columns, or the Microsoft Excel Add-in can map fields automatically. For automatic mapping to succeed, the column headings in your worksheet must be identical to the field names in ALM.
- When you upload a new or existing requirement, test, or defect to ALM, it is created with the user name that you use to upload it. For example, if you upload a defect to ALM with alex_alm as the user name, the defect's history shows alex_alm as the person who changed the defect.
- You cannot upload the following ALM fields from Excel:

Module Name	Field Name
Requirements	Date, Time, Modified, ReqID, Direct Cover Status
Test Plan	Modified, Test Step Actual, Test Step Exec Date, Test Step Exec Time, Test Step Source ID, Status
Defects	Defect ID, Modified

- You must have full create and modify permissions in ALM for the type of record that you are uploading. For example, if you are uploading tests, you must have permissions for adding and modifying the test plan tree, design steps, and folders.

Version Control Considerations

You can upload requirements or tests from Microsoft Excel to a version control enabled project in ALM.

Consider the following when uploading to a version control enabled project:

- When you upload a new requirement or test, it is created in ALM, with a checked in status.
- When you upload an existing requirement or test that is checked in, it is automatically checked out and then checked in during upload. ALM adds a comment to the version history specifying that the requirement or test was updated by upload from Excel.
- If you upload an existing requirement or test that is checked out by the same user that you use to upload the data, the data is uploaded. If the requirement or test is checked out by another user, the upload fails and an error message is displayed.

For more information on version control, refer to the *Application Lifecycle Management User Guide*.

Logging in to ALM

You must log in to your ALM server before mapping, validating, or uploading your data.

To log in to ALM:

1. Click **Login**. The ALM Login dialog box opens.

Enter your ALM server and the virtual directory name (http://<ALM server name>[:<port number>]/qcbn) and click **Authenticate**.

2. Enter your email address and click **Continue**.

You are redirected to log in using one of the following options:

- Your corporate credentials (at a URL belonging to your corporate domain). The credentials may be a username and a password, or an authentication certificate.

If your company set up Single-Sign-On with ALM and you already logged in to another application with your corporate credentials, you may be redirected to the ALM application without needing to log in again.

- Your Micro Focus SaaS account password.



Note: The number of times you can enter incorrect credentials and the length of time you can remain logged in are managed outside of ALM. This is managed at the corporate level or by Micro Focus SaaS.

3. In the ALM Login dialog box, select your domain and project and click **Login**.

Mapping Your Data

You can manually map ALM fields to worksheet columns, or the Microsoft Excel Add-in can automatically map them.

To map your data:

1. Select the cells in your Excel worksheet that contain the data that you want to upload. Do not include the column headings row in your selection.
2. Click **Mapping**. The Field Mapping dialog box opens.
3. Click **Open** to open the file explorer to search for existing mapping files to use for mapping your Excel data. The file contains the entity type and the field mapping to use.
4. Select the entity type (Requirements, Defects, or Tests) to which you are mapping your data. The ALM fields for the selected entity type appear in the lower left pane. The required fields appear in red in the lower right pane.



Note: If you select Requirements, you can apply a requirement type to assign to all requirements you are uploading.

5. Map your Excel columns to the ALM fields:

- **Manually.** Move the non-required ALM fields to which you want to map from the left pane to the right pane, and then enter the Excel column letter that corresponds to each ALM field.
- **Automatically.** Click **Auto-map** and enter the column headings row number.

Note: The column headings must exactly match the ALM field names.

6. Unselect **Add or update ALM field label as header to Excel** if you do not want to create a header row in Excel.

Note: If you click **Auto-map**, you will be prompted to unselect **Add or update ALM field label as header to Excel**.

7. Click **Save** if you want to save your field mapping for future use.

8. Click **OK** to apply your mapping to your Excel worksheet.

Validating Your Data

Once you have mapped your data to the ALM fields, you must check that your data is valid for its mapped field. For example, if the ALM field allows only certain values, your data cannot contain other values.

To validate your data:

1. Select the cells that contain your data and click **Validate**. Do not include the column headings row in your selection.

The ALM Microsoft Excel Add-in validates your data against the requirements for each ALM field. Any invalid cells are marked in red, and an error message is displayed when you put your cursor on the cell.

2. Correct any invalid cells and click **Validate** again.

Note: If you subsequently add or modify your data, you can validate again, selecting only the modified or added cells or selecting all the cells. If you do not validate prior to uploading your data, the ALM Microsoft Excel Add-in validates your data before uploading.

Uploading Data to ALM

You upload data to ALM from your Excel worksheet.

To upload your data:

1. Click **Upload to ALM**. The data is uploaded to ALM.

Note: If you click **Upload to ALM** without first clicking **Validate**, the ALM Microsoft Excel Add-in validates your data before uploading. Any invalid cells are marked in red, and an error message is displayed when you put your cursor on the cell. If there are no errors, the data is uploaded to ALM.

2. Refresh the view in ALM. Review the uploaded data and make the necessary additions and adjustments.

Chapter 5: Limitations

- **Logging in with Special Characters:** Users cannot log in to the Microsoft Excel Add-in if the user name contains an ampersand. This is because the REST API and applications that use REST, such as the Microsoft Excel Add-in, do not support user names with an ampersand, due to a LWSSO technology limitation.
- **Multiple Value Fields:** The following limitation occurs when you are exporting data to an ALM field of type Lookup List, for which multiple values are allowed, and for which ALM verifies the value. If you try to export data to such a field for a record containing duplicate values (for example, High; High), the export does not complete successfully but a new record is created in ALM with empty field values.
- **Using a Transport Layer Security (TLS) secure connection when connecting to the ALM server:** To use the TLS version 1.2 secure connection for the Excel add-in, install .NET 4.5 on your client machine. (Using a TLS version 1.2 connection with .NET 4.0 causes an “Unable to connect to the ALM server” error.)

